

Simple Regression Model

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Roadmap

Ordinary Least Squares Estimation

Population Regression Function

We currently consider the statistical relationship between two variables X and Y

To demonstrate important concepts, let's focus on the example given by the textbook:

How do student-teacher ratios affect student performance?

Before we get technical, let's first think about the content of that question; i.e., what is the hypothesized relationship b/w the two variables?

(It's always a good idea to first think about your econometric model intuitively)

The primitive argument goes like this:

The fewer students there are per teacher,
the more individualized instruction can be

Individualized instruction helps students

We would hypothesize that a lower student-teacher ratio, all else
equal, should have a positive effect on student performance

It's a simple argument, really

Now, turning technical, we believe there is a functional relationship between student-teacher ratio and student performance:

$$TestScore = f(STR, u),$$

where

- *TestScore* captures student performance
- *STR* is the student-teacher ratio
- *u* captures all other things that explain *TestScore* (over and above *STR*)
Examples: intelligence, luck

In this example, the three different variables can be given generic names

- *TestScore* is called the dependent variable
- *STR* is called the independent variable or explanatory variable or regressor
- u is called the error term

We hypothesize a negative relationship b/w *TestScore* and *STR*,
formally: $\partial TestScore / \partial STR < 0$

The equation

$$TestScore = f(STR, u),$$

together with the hypothesis

$$\frac{\partial TestScore}{\partial STR} < 0$$

summarize an *econometric model*

(a way an economist would think of a relationship between two variables statistically)

Digression: why do we need to include u ?

If we did not include u as part of the function $f(\cdot)$ then we would presume that the relationship b/w *TestScore* and *STR* was *deterministic*

It would mean that once we know *STR* we also know *TestScore*

This is almost like saying that they are one and the same thing

Deterministic relationships often make sense in the natural sciences,
example: relationship b/w Celsius and Fahrenheit

In economics, relationships b/w variables are never deterministic
but are subject to some degree of randomness and the presence of
the *error term* u allows for that

So we think there is a functional relationship between *TestScore* and *STR*

Problem: which function should $f(\cdot)$ be?

There are infinitely many possibilities!

There is no way of knowing

We will now make three important simplifying assumptions regarding the functional form of $f(\cdot)$

1. $f(X, u)$ is additively separable in X and u :

$$f(X, u) = g(X) + h(u)$$

2. $g(X)$ is a linear function:

$$g(X) = \beta_0 + \beta_1 X,$$

where β_0 and β_1 are called *coefficients* of the model

3. $h(u)$ is the simple identity function:

$$h(u) = u$$

Combining these three results in

$$f(X, u) = \beta_0 + \beta_1 X + u$$

The function $g(X) = \beta_0 + \beta_1 X$ plays an important role

It has its own name:

Definition

$g(X)$ is called the **population regression function (PRF)**.

The coefficients β_0 and β_1 are unknown

In a sense, we still don't really know the PRF because we do not know β_0 and β_1

But at least we know the principal class of the PRF
(it's linear in the coefficients)

But we will see soon that this is not so bad,
there is a good way of estimating the coefficients

Applying $f(X, u)$ to our example results in

$$TestScore = \beta_0 + \beta_1 STR + u,$$

Our hypothesis $\frac{\partial TestScore}{\partial STR} < 0$ from earlier simply translates to $\beta_1 < 0$

Our econometric model therefore is summarized in one line:

$$TestScore = \beta_0 + \beta_1 STR + u, \text{ with } \beta_1 < 0$$

Our main obsession this semester will be to learn how to estimate the coefficients β_0 and β_1 and study the characteristics of these estimates

But let's suppose, for the sake of illustration, that an oracle was so kind to tell us that $\beta_0 = 700$ and $\beta_1 = -2.50$

What would we make of this information?

Let's graph the PRF



β_0 is the intercept, β_1 is the slope

A closer look at the PRF

Claim:

The PRF tells us the *expected* TestScore for a person with a given value of STR

More technically, this is called the *conditional expectation of TestScore given STR*

Let's prove that claim

Recall: $TestScore = \beta_0 + \beta_1 STR + u$

The expected $TestScore$ given STR is given by the conditional expectation $E[TestScore|STR]$

Plugging in the top equation and then solving results in

$$\begin{aligned} E[TestScore|STR] &= E[\beta_0 + \beta_1 STR + u|STR] \\ &= \beta_0 + \beta_1 E[STR|STR] + E[u|STR] \\ &= \beta_0 + \beta_1 STR + 0 \\ &= g(STR) \end{aligned}$$

The lhs is the expected $TestScore$ for a person with a given value of STR ; the rhs is the PRF; therefore we have just proved the claim

In the previous calculations we have made one important assumption, namely

$$E[u|STR] = 0$$

This is the so-called **conditional mean independence assumption**

We now need to look at conditional mean independence from two angles:

1. mathematical
2. intuitive

The mathematical definition is straightforward

Definition

Two random variables X and Y are **conditionally mean independent (CMI)** if

$$E[Y|X] = E[Y].$$

Corollary

If X and Y are CMI then $E[X|Y] = E[X]$.

Corollary

If X and Y are CMI then $\sigma_{XY} = \rho_{XY} = 0$.

On the previous slide, X and Y played the role of generic random variables

Applying the CMI definition to u and STR would mean, strictly speaking, that we require

$$E[u|STR] = E[u]$$

in order for u and STR to be CMI

But we wrote earlier that $E[u|STR] = 0$

Reconciliation: without loss of generality we set $E[u] = 0$

Offering intuition for CMI

Example: average daytime maximum temperature

- in Canberra throughout the year is 20 degrees Celsius
- in Canberra in summer is 27 degrees Celsius
- in Canberra in winter is 13 degrees Celsius

Mathematically:

$$E[Temp] = 20 \quad E[Temp|summer] = 27 \quad E[Temp|winter] = 13$$

and therefore

$$E[Temp] \neq E[Temp|summer] \quad \text{and} \quad E[Temp] \neq E[Temp|winter]$$

Temperature is not mean independent of season

More examples:

- $E[\text{Intelligence}|\text{Education}]$
- $E[\text{Intelligence}|\text{identify as male}]$
- $E[\text{Height}|\text{identify as female}]$

Which ones are conditionally mean independent?

Intuitively, when are u and STR CMI?

In analogy to the previous examples, they would be CMI if STR is not predictive of u

It seems obvious that if u were a true random error, then STR is not predictive of u and they would therefore be CMI

Remember, earlier we said that u may be capturing things such as intelligence or luck

Do you believe that STR is predictive of these?

Let's say I told you that when I went to high-school (many many years ago) I had 28 class mates (including myself)

Then the econometric model would assign the following *expected TestScore* for me:

$$Testscore = 700 - 2.5 \cdot 28 = 630$$

This is merely the *expected TestScore*

Actual *TestScores* for individual students most likely differ

In any class of 28 students, some students will do better and some will do worse for various reasons unrelated to the student-teacher ratio

In other words, it is not deterministic that every student in a class of 28 will receive a test score of 630 (obvious, isn't it!?)

The role of the error term u is to bring in some randomness across students

Actual TestScore (as opposed to expected) is as follows:

$$TestScore = 700 - 2.5 \cdot 28 + u = 630 + u$$

The way to think about it is this:

- Each student in my class of 28 students starts out on a basis of 630 (that is explained through the effect of student-teacher ratio on *TestScore*)
- On top of that each student has her/his own error term u that shifts her/him further up or down

Consider three of my class mates

Student	STR	u
A	28	20
B	28	-15
C	28	0

How would we compare students A and B?

- Student A has a test score of 650, student B of 615
- Reasons: student A is cleverer and/or is a better test taker and/or was more lucky on test taking day
- These reasons are unrelated to STR and influence $TestScore$ as well

What important role does student C play here?

Student C is the average student, with a score of 630

We see that the econometric model can make predictions for the *average student*

In the absence of any other influences on *TestScore*, the model predicts a score of 630

This offers us a new view of the function $f(\cdot)$

When we write the model as $TestScore = \beta_0 + \beta_1 STR + u$
we make an assumption about how we think about the average student

Put differently we have an econometric model about the *expected* test score:

$$E[TestScore|STR] = \beta_0 + \beta_1 STR$$

So we are not actually studying

- how *STR* determines *TestScore*
but instead
- how *STR* determines *expected TestScore*

This is an important difference!